Mooney M20J 201 (M20J)

Lycoming IO-360-A3B6D, 200 hp · 4 seats · Typical M20J 201 (1977–1998, 2,740 lb) POH procedures; flap limit speed and some gear speeds changed by model year.

V-speeds & limits

Vr	**60–65 KIAS (rotate)**
Vx	**approx 66–70 KIAS**
Vy	**91 KIAS (sea level, 2,740 lb)**
Best glide	**91 KIAS (2,740 lb, gear & flaps up)**
Vfe	**110–115 KIAS (varies by model year)**
Vlo	**132 KIAS extend / 107 KIAS retract**
Vle	**165 KIAS**
Va	**112 KIAS (2,740 lb)**
Vno	**174 KIAS**
Vne	**195 KIAS**
Vs / Vso	**approx 61 / 55 KIAS (max weight)**
Vref	**70–75 KIAS final (full flaps); 65–70 KIAS over threshold**

Normal procedures

Before Start

Preflight inspection	**COMPLETE**
Passenger briefing	**COMPLETE (belts, door – both latches, exits, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door	**CLOSED & LATCHED (both latches); baggage door LOCKED**
Parking brake	**SET**
Avionics master	**OFF**
Landing gear switch	**DOWN**
Fuel selector	**FULLEST TANK (LEFT or RIGHT – OFF is a detented position, avoid it)**
Fuel quantity	**CHECK both gauges**
Cowl flaps	**OPEN**
Ram air	**CLOSED (if fitted)**
Alternate static source	**OFF (normal)**
Master switch	**ON**
Landing gear indicator	**GREEN (down)**
Annunciators / gear warning	**TEST**
Beacon / strobes	**ON**

Engine Start

Propeller area	**CLEAR**
Mixture	**FULL RICH**
Propeller	**HIGH RPM (full forward)**
Throttle	**1/4 OPEN**
Fuel boost pump	**ON 3–5 sec until fuel pressure / flow indicates (prime), then OFF**

▶ Engine hot or flooded?

No:

Yes:

Fuel boost pump	**OFF (do not prime)**
Mixture	**IDLE CUTOFF if flooded**
Throttle	**1/2 OPEN – advance mixture to FULL RICH and retard throttle as engine fires**
Magneto / starter switch	**START (release when engine fires)**
Throttle	**1000 RPM**
Oil pressure	**25 psi within 30 sec (normal 60–90 psi)**
Alternator	**CHECK charging (ammeter / voltmeter)**
Avionics master	**ON**
Engine instruments	**CHECK**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeter, HSI / DG, turn coordinator during turns)**
Avionics / GPS	**SET**
Autopilot / electric trim	**CHECK & OFF**
Elevator trim	**SET takeoff**
Fuel selector	**CHECK fullest tank (LEFT or RIGHT)**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH (lean for smooth running above approx 5,000 ft elevation)**
Throttle	**2000 RPM**
Magnetos	**CHECK L & R (175 RPM max drop, 50 RPM max difference)**
Propeller	**CYCLE 2–3 times (RPM drops, then returns)**
Engine instruments	**CHECK (oil pressure 60–90 psi, oil temp green, fuel pressure green)**
Alternator	**CHECK charging**
Suction	**CHECK in green**
Throttle	**IDLE check (approx 600–700 RPM), then 1000 RPM**
Fuel selector	**FULLEST TANK (feel for detent)**
Fuel boost pump	**ON**
Cowl flaps	**OPEN**
Ram air	**CLOSED**
Propeller	**HIGH RPM**
Flaps	**TAKEOFF (15°) or UP (check indicator)**
Trim	**SET takeoff**
Landing gear switch	**DOWN – CHECK**
Pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Cabin door / windows	**LATCHED (both door latches)**
Takeoff briefing	**COMPLETE (abort point; gear up below 107 KIAS with positive rate)**

After Takeoff / Climb

Landing gear	**UP (positive rate, no usable runway, below 107 KIAS)**
Gear indicator	**GREEN OUT, gear unsafe light OUT**
Flaps	**UP (above 80 KIAS)**
Fuel boost pump	**OFF (check fuel pressure steady)**
Climb power	**FULL THROTTLE / 2700 RPM; cruise climb 25 in Hg / 2500 RPM**
Mixture	**LEAN above approx 5,000 ft (EGT / smooth running)**
Cowl flaps	**OPEN as required (CHT / oil temp in green)**
Airspeed	**100–105 KIAS (cruise climb)**
Engine instruments	**CHECK**

Cruise

Cruise power	**SET (2300–2500 RPM, 65–75% power typical)**
Mixture	**LEAN per POH / EGT**
Cowl flaps	**CLOSED (as required)**
Ram air	**OPEN in clear air (CLOSED in rain, icing or dust)**
Fuel selector	**ALTERNATE LEFT / RIGHT to keep tanks balanced – note time**
Fuel quantity / pressure	**CHECK against plan**
Engine instruments	**CHECK (oil temp 245°F max)**
Altimeter / heading	**CHECK**

Descent

Altimeter	**SET**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Mixture	**ENRICH progressively**
Power	**REDUCE gradually (avoid rapid cooling)**
Ram air	**CLOSED**
Cowl flaps	**CLOSED**
Speed control	**PLAN EARLY – the 201 does not slow quickly; level off and reduce power before the pattern to be below 132 KIAS for gear**
Approach briefing	**COMPLETE**
Seat belts / harnesses	**SECURE**

Before Landing

Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Fuel boost pump	**ON**
Mixture	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 132 KIAS (Vlo extend)**
Landing gear	**DOWN – GREEN light, gear unsafe light OUT (gear may be used as a speed brake; Vle 165 KIAS)**
Gear visual indicator (floor)	**CHECK marks aligned – gear down**
Flaps	**TAKEOFF below 110–115 KIAS; FULL on final**
Propeller	**HIGH RPM (on final)**
Airspeed	**70–75 KIAS final; 65–70 KIAS over threshold (lower at light weight)**
Landing light	**ON**
Landing gear	**GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (identify the flap switch before moving)**
Cowl flaps	**OPEN**
Fuel boost pump	**OFF**
Pitot heat	**OFF**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trim	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttle	**1000 RPM (cool-down)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Mixture	**IDLE CUTOFF**
Magnetos	**OFF (after propeller stops)**
Master switch	**OFF**
Control lock	**INSTALL**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

Alternator warning light, ammeter discharge or voltage below limits

Alternator (field) switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

▶ Alternator restored?

Yes:

Ammeter / voltmeter ..... **MONITOR**

No:

Alternator switch ..... **OFF**

Non-essential electrical load ..... **REDUCE (battery only, limited endurance)**

Landing gear ..... **PLAN early extension while battery power remains (manual extension if needed)**

Deferred: Alternator failed – battery power only; extend gear early, land as soon as practical. ..... **NOTED**

Low Oil Pressure

Oil pressure below 25 psi in flight

Oil pressure / oil temperature ..... **CHECK and cross-check**

▶ Oil temperature rising or pressure falling further?

Yes – genuine:

Power ..... **REDUCE to minimum required**

Airport ..... **DIVERT to nearest suitable – stay within glide of a landing area**

Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

Flight ..... **CONTINUE, monitor oil temperature and pressure**

Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

High Oil / Cylinder Head Temperature

Cowl flaps ..... **OPEN**

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower pitch attitude)**

Power ..... **REDUCE**

▶ Temperature returning to green?

Yes:

Flight ..... **CONTINUE, monitor**

No:

Oil pressure ..... **MONITOR for drop**

Deferred: High engine temperature – land as soon as practical, cooling / baffling check required. ..... **NOTED**

Rough Running / Low or Fluctuating Fuel Pressure

Fuel boost pump ..... **ON**

Fuel selector ..... **SWITCH TANKS**

Mixture ..... **ADJUST for smooth running**

Ram air ..... **CLOSED**

Magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (CHT, EGT, fuel pressure)**

▶ Engine running smoothly?

Yes:

Deferred: Rough running cleared – leave boost pump ON; report to maintenance. ..... **NOTED**

No:

Power ..... **REDUCE to smoothest setting**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Flap Failure / Asymmetry

Flap switch ..... **NEUTRAL / stop – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – flaps up approach 75–80 KIAS, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Landing Gear Fails to Retract / Extend Normally

Airspeed ..... **BELOW 107 KIAS to retract / BELOW 132 KIAS to extend**

Landing gear actuator circuit breaker ..... **CHECK / RESET once**

Landing gear switch ..... **CYCLE once to the desired position**

Gear warning horn / lights ..... **CHECK (bulb or horn fault vs real gear fault)**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM (floor visual indicator)**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear switch ..... **DOWN – leave down**

Airspeed ..... **BELOW 165 KIAS (Vle)**

Deferred: Gear will not retract – gear down, reduced performance and range, land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Emergency Gear Extension (Manual).

Cabin Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 90 KIAS**

Belts / loose items ..... **SECURE**

▶ Attempt to close?

Yes – passenger can assist:

Pilot storm window ..... **OPEN (equalise cabin pressure)**

Door ..... **PULL CLOSED and engage both latches**

No:

The airplane flies normally with the door trailing open; expect noise and buffet.

Deferred: Cabin door unlatched – return and land as soon as practical, normal approach speed. ..... **NOTED**

Vacuum / Attitude Indicator Failure

Suction gauge ..... **CHECK**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**

Failed instruments ..... **COVER**

Autopilot ..... **OFF if coupled to failed gyro**

IMC ..... **EXIT / AVOID**

Deferred: Vacuum / gyro failure – partial panel; prefer VMC approach. ..... **NOTED**

Propeller Overspeed

RPM above 2700 and not responding to prop lever

Throttle ..... **REDUCE**

Propeller ..... **LOW RPM (full aft)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor is oil-operated)**

Power ..... **SET minimum that keeps RPM at or below 2700**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Throttle (memory) ..... **CLOSED**

Brakes (memory) ..... **AS REQUIRED – stop straight ahead**

Mixture ..... **IDLE CUTOFF**

Fuel selector ..... **OFF**

Magnetos ..... **OFF**

Master switch ..... **OFF**

Engine Failure After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

Airspeed (memory) ..... **80–91 KIAS (91 KIAS gear & flaps up)**

▶ Runway remaining to land on?

Yes:

- Landing gear ..... **DOWN (if retracted, below 132 KIAS)**
- Flaps ..... **AS REQUIRED**
- Land ..... **STRAIGHT AHEAD**

No – land ahead off-field:

- Landing gear ..... **DOWN on firm surface / UP on rough terrain or water**
- Flaps ..... **FULL when landing assured**
- Fuel selector ..... **OFF**
- Mixture ..... **IDLE CUTOFF**
- Magnetos ..... **OFF**
- Master switch ..... **OFF (after gear and flaps set)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**91 KIAS (best glide, gear & flaps up)**
Fuel selector (memory)	**SWITCH TANKS (feel for detent)**
Fuel boost pump (memory)	**ON**
Mixture (memory)	**FULL RICH, then lean as required**
Magnetos (memory)	**BOTH – check**
Alternate air (memory)	**ON (if manual control fitted; door is otherwise automatic)**
Propeller (memory)	**LOW RPM (full aft) to extend glide**

▶ **Restart attempt practicable?**

Yes – restart:

Throttle	**1/4 OPEN**
Starter	**ENGAGE if propeller has stopped**

▶ **Engine restarted?**

Yes:

Power	**AS REQUIRED**
Fuel boost pump	**ON**

Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel system. **NOTED**

No:

Continue with forced landing.

No – forced landing:

Forced landing without power:

Seats / belts / harnesses	**SECURE**
Landing gear	**DOWN on firm surface (below 132 KIAS) / UP on rough terrain or water**
Flaps	**FULL when landing assured**
Airspeed	**70–75 KIAS on final**
Fuel selector	**OFF**
Mixture	**IDLE CUTOFF**
Magnetos	**OFF**
Master switch	**OFF (after gear and flaps set)**
Cabin door	**UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	**OFF**
Mixture (memory)	**IDLE CUTOFF**
Fuel boost pump (memory)	**OFF**
Throttle (memory)	**CLOSED**
Cabin heat / defroster / vents	**OFF / CLOSED**
Airspeed	**INCREASE (120–130 KIAS) to help extinguish fire**
Landing gear	**DOWN electrically below 132 KIAS before switching master off (or manual extension)**
Forced landing	**EXECUTE (do not attempt restart)**

Engine Fire During Start

Starter (memory) ..... **CONTINUE CRANKING**

▶ Engine starts?

Yes:

Throttle ..... **1700 RPM for a few minutes**

Engine ..... **SHUT DOWN and inspect for damage**

No:

Mixture ..... **IDLE CUTOFF**

Throttle ..... **FULL OPEN**

Fuel boost pump ..... **OFF**

Fuel selector ..... **OFF**

Starter ..... **CONTINUE CRANKING to draw fire into engine**

Magnetos / master switch ..... **OFF**

Fire extinguisher ..... **OBTAIN / USE**

Aircraft ..... **EVACUATE**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

Master switch (memory) ..... **OFF**

Avionics master (memory) ..... **OFF**

Cabin heat / vents ..... **CLOSED**

Fire extinguisher ..... **ACTIVATE (then ventilate cabin)**

▶ Fire out and electrical power required?

Yes:

Circuit breakers ..... **CHECK – do not reset faulted**

Master switch ..... **ON**

Avionics / equipment ..... **ON one at a time, check for smoke**

Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

No:

Landing gear and flaps are electric: plan a manual gear extension (see Emergency Gear Extension) and a flaps-up landing.

Deferred: Electrical power off – manual gear extension, flaps-up landing, no landing light. ..... **NOTED**

Emergency Descent

Throttle (memory) ..... **CLOSED**

Propeller (memory) ..... **HIGH RPM**

Airspeed (memory) ..... **BELOW 132 KIAS, then landing gear DOWN**

Flaps (memory) ..... **UP**

Airspeed (memory) ..... **165 KIAS gear down (smooth air) / 112 KIAS (turbulence)**

Mixture ..... **ENRICH as required**

Emergency Gear Extension (Manual)

Gear does not extend electrically

Airspeed	**BELOW 132 KIAS (slower makes the lever easier to operate)**
Landing gear actuator circuit breaker	**PULL**
Landing gear switch	**DOWN**
Manual gear extension lever (red, floor beside pilot seat)	**UNLATCH – pull release / T-handle to disengage the actuator clutch**
Manual gear extension lever	**PULL fully back and return – REPEAT (approx 12–20 strokes) until lever stops and green light illuminates**
Landing gear indicator	**GREEN; gear unsafe light OUT; floor visual indicator ALIGNED**
Manual lever	**RETURN to stowed position and LATCH**

Do not reset the gear actuator circuit breaker or move the gear switch in flight after a manual extension. Leave the gear down.

▶ Green light and visual indicator obtained?

Yes:

Deferred: Gear extended manually – do not retract; gear system inspection required. ..... **NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, door exit)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**FULL**
Approach	**NORMAL, 70–75 KIAS**
Fuel boost pump	**OFF**
Cabin door	**UNLATCHED**

On short final:

Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Magnetos	**OFF**
Master switch	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**